GRANT LOCKE

Boston, MA / Seattle, WA | 425-559-1886 | GLocke99@bu.edu | linkedin.com/in/grant-locke425
U.S. Citizen | Available full-time June 2027

Mechanical engineering senior focused on structural analysis and simulation (ANSYS, Abaqus, COMSOL, OpenRocket), backed by hands-on design, GD&T documentation, and inspection of production hardware.

EDUCATION

Boston University, College of Engineering | Boston, MA — Expected May 2027
B.S. Mechanical Engineering | Technology Innovation Concentration
Relevant Coursework: Mechanics of Materials, Thermodynamics, Heat Transfer, Fluid Mechanics, Robotics, Manufacturing Systems, CAD & Engineering Design, Engineering Data Analysis

TECHNICAL SKILLS

Analysis & Simulation: ANSYS, Abaqus, COMSOL Multiphysics (structural FEA, meshing strategy, mesh convergence, boundary conditions), OpenRocket flight simulation, MATLAB, Simulink, Python, engineering data analysis
Design & CAD: Creo Parametric, SolidWorks (CSWA), Siemens NX, AutoCAD, Windchill (PDM/release), mechanism design, assemblies, detailed drawings, GD&T, tolerance stack-ups, BOM/MBOM, DFM/DFA
Inspection & Test: Inspection planning, CMM/metrology evaluation, prototype fit checks, load testing, sensors, Arduino, soldering
Fabrication: Tooling/fixture design, CNC/manual machining, FDM/SLA 3D printing, mechanical assembly, Lean/5S, standard work

ENGINEERING EXPERIENCE

General Dynamics Ordnance and Tactical Systems | Bothell, WA — Jun 2026 – Aug 2026
Manufacturing Engineering Intern

- Redesigned a rotating loading fixture in Creo for 70-lb hardware, developing bearing/shaft supports, indexed positioning, locking features, and assembly interfaces for repeatable, operator-accessible use.
- Created and revised ~8 controlled tooling drawings with GD&T and released them through Windchill, incorporating machinist feedback on fabrication, fit, and assembly.
- Translated drawings and GD&T into an inspection plan covering ~200 dimensional and geometric characteristics (diameters, depths, runout, threads, surface finish).
- Ran a trade study of 5 CMM and shop-floor metrology solutions against accuracy, measurement volume, deep-feature access, operating environment, implementation risk, and cost.
- Sourced gages, bore-measurement systems, and specialty fixtures with Quality, Procurement, and suppliers, trading technical capability against cost and lead time.
- Designed PLA/TPU production aids and a point-of-use shadow-board system for ~20 users; revised fabrication work instructions and supported hardware builds and assembly.

PACCAR – Kenworth Truck Company | Renton, WA — May 2026 – Jun 2026
Manufacturing Engineering Intern

- Designed a robotic paint-cart concept in Creo around mobility, material presentation, operator access, and production-space constraints.
- Conducted Gemba walks, ergonomic/safety assessments, and standard-work reviews; ranked risks, investigated causes, and presented corrective actions to manufacturing leadership.

Sigma Defense Systems | Perry, GA — Dec 2025 – Jan 2026
Mechanical Engineering Intern

- Performed ANSYS FEA and tolerance checks on transit-case components and protective foam layouts for defense equipment, balancing fit, protection, and manufacturability.
- Designed the components in SolidWorks and validated them through prototype fit checks and assembly, folding supplier feedback into iterative design revisions.

WillScot | Seattle, WA — May 2025 – Sep 2025
Project Engineering Intern

- Updated AutoCAD layouts, installation drawings, and shipping documentation for modular classrooms; verified site requirements and resolved layout issues with project and field teams.

Boston University EPIC Engineering Shop | Boston, MA — Sep 2024 – May 2025
Engineering Shop Student Assistant

- Fabricated and supported prototypes using CNC milling, manual lathe, laser cutting, and FDM/SLA printing; troubleshot student builds for manufacturability, fixturing, and tolerance issues.

SELECTED ENGINEERING PROJECTS

Mars Rover Team – University Rover Challenge | Boston University In Progress

55+ member team building a Mars rover for the Mars Society's University Rover Challenge; member of two mission sub-teams.

Equipment Servicing – Gripper / End Effector Design and Analysis

- Own the design calculations for the rover arm's servo-actuated gripper: grip force required to prevent slip, actuator torque across the mechanism's range of motion, and loads on fingers, pins, brackets, and the mounting interface.
- Running FEA on critical components to evaluate stress and deformation and checking results against hand calculations; analyzing clearances, tolerances, and interference to prevent binding and unwanted play.
- Designing gripper linkages and joints in SolidWorks; selecting manufacturing methods by component load and geometry, prototyping to verify fit and motion, and designing wear parts for fast replacement in competition.
- Implementing open/close/stop commands, travel limits, and stall or failed-grasp detection from jaw position or motor current with the electrical and software teams; writing Python scripts to log, plot, and compare commanded vs. measured performance.
- Developing a repeatable subsystem test plan for grip retention, opening range, repeatability, and cycle time across representative objects and arm orientations; reporting gripper mass and CG to the arm team for joint-load and payload budgets.
- Maintaining drawings, an accurate parts list, and assembly and maintenance instructions; inspecting critical dimensions on prototype parts and documenting design changes.

Autonomous Traversal – Navigation Sensor Integration

- Designing CAD mounting brackets and support structures for navigation cameras and sensors that interface with the chassis, preserve a clear terrain view, and allow alignment, adjustment, and maintenance.
- Sizing mounts for weight, stiffness, vibration, and rough-terrain loads; selecting materials, fasteners, and manufacturing methods with protected cable routing and connector access.
- Coordinating sensor placement and orientation with electrical and software teams; planning field tests to verify mount alignment, retention, and sensor field of view.

MBTA Rail Switch Gap Measurement | Senior Design, six-person team In Progress

- Defining point-to-stock rail gap measurement requirements for switch-position monitoring and maintenance detection.
- Assessing sensor survivability against vibration, shock, debris, moisture, temperature variation, and repeated actuation.
- Developing a prototype and test scope to evaluate gap measurement in a simulated rail environment.

High-Power Model Rocket Design and Flight Simulation

- Modeled a LOC-IV high-power rocket in SolidWorks, defining airframe geometry, component packaging, and payload layout.
- Ran OpenRocket flight simulations to evaluate motor selection, stability, and predicted flight profile, connecting aerodynamic and packaging choices to predicted performance.

Structural Truss Design and Analysis

- Evaluated acrylic truss member forces, stress, and buckling limits in MATLAB and ANSYS, verified by hand calculations using joint equilibrium.
- Led a four-person team through fabrication, load testing, and model-to-test comparison, trading load capacity against material and joint cost.

2.5-DOF Electromechanical Mechanism

- Designed and built a motor-driven carriage and rack-and-pinion launcher from machined and 3D-printed parts; refined alignment, clearances, and timing through iterative testing.

Bluetooth Doorbell Prototype

- Modeled a SolidWorks enclosure integrating a gesture-triggered, servo-actuated chime with wiring access and assembly clearances.

Temperature Monitoring and Alarm System

- Built an Arduino system with LCD, buzzer, and relay that displays readings and triggers an alarm outside a 70–75 °F operating range.

ADDITIONAL EXPERIENCE

Art Legacy | Project Management Consultant Summer 2023

- Identified warehouse and shipping bottlenecks; used takt time analysis to recommend layout and staffing changes and built Excel trackers for operating and financial decisions.

Brooklyn Brothers Pizza | Assistant Manager May 2024 – Sep 2024

- Led a 10-person team, managing scheduling, training, and workflows for high-volume service.